

RIVER ROUTING

GLY606 Water Data Analysis & Modeling

Oct 25th 2024



Directly go to
<https://jupyterhub.cuahsi.org/>

Let's use “**WRF-Hydro
v5.1 – NWM v2.0**”
environment

Server Options

- **Python v3.9.7 - JupyterLab Interface**

This is a general-purpose scientific computing environment built on Python version 3.9.7. When using this server, you will be working through the JupyterLab interface. It comes with pre-configured, widely used scientific libraries including scipy, pandas, geopandas, sympy, scikit-learn, matplotlib, ulmo, landlab, gdal, xarray, TauDEM, hscient, dataretrieval, rasterio, pyproj, fiona, geemap, fsspec, and others.

- **R v4.2.3 - JupyterLab Interface**

This serves as a general-purpose scientific computing environment built on the R programming language. When using this server, you will have access to both the JupyterLab and RStudio interfaces. The environment is pre-configured with commonly used tools and libraries, including randomForest, forecast, sf, rjsonio, ncd4, ggmap, dataRetrieval, and WaterML.

- **SUMMA Modeling**

This is a computing environment built on Python version 3.9 and pre-installed tools to support using the Structure for Unifying Multiple Modeling Alternative (SUMMA). SUMMA is a hydrologic modeling framework that can be used to configure a wide range of hydrological model analysis. When using this server, you will be working through the JupyterLab interface. The environment comes with limited pre-installed scientific Python packages. For more information on the SUMMA model, visit the project [webpage](#).

- **WRF-Hydro v5.1 - NWM v2.0**

This computing environment is structured around Python version 3.9 and is tailored for seamless operation with the WRF-Hydro model because it has the source code pre-compiled. WRF-Hydro is a community hydrological model suitable for a range of projects, including real-time flash flood prediction, long-term water supply projections, and evaluations of regional hydroclimate impacts. Notably, WRF-Hydro has been used as the core hydrological module within the National Water Model (NWM) versions 2.0 and 2.1. When using this server, you will be working through the JupyterLab interface. While the environment features a curated selection of pre-installed scientific Python packages, users can follow CUAHSI's guideline to create a customized conda environment, and then install specific Python packages that cater to their unique project requirements. For more information on the WRF-Hydro model, visit the project [website](#).

Homework #6

- Form a team of 2 to 3 members
- Each team will only need to submit one homework
 - Pick one river basin (medium size, basin area ranges between 1000 and 4000 km²)
 - Find all available USGS sites available in the basin.
 - Plot a domain map for this basin with the location of all USGS sites on it
 - Conduct exploratory data analysis (EDA) using the USGS data
 - Write a two-page report
 - 1. Briefly introduce this river basin and why does your team pick this river basin?
 - 2. Summarize the EDA (Did you find any interesting trend in mean annual flow or peak flow?)
 - Summarize the information for all USGS sites into a table and attach it in the appendix (This information is not counted towards the two-page limit)
 - Information required for each site: site name, site ID, latitude, longitude, confluence area, data availability (start year, end year), etc.

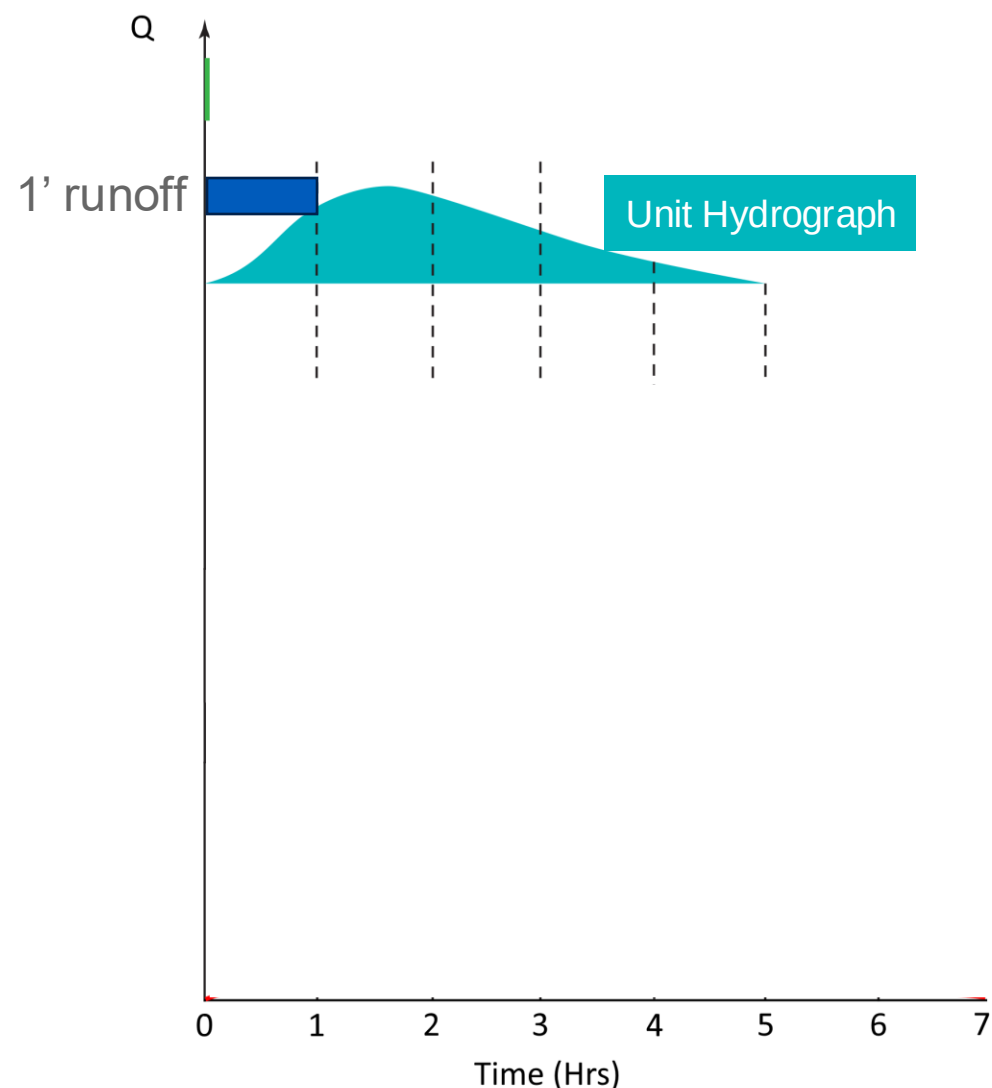


Brief introduction to river routing

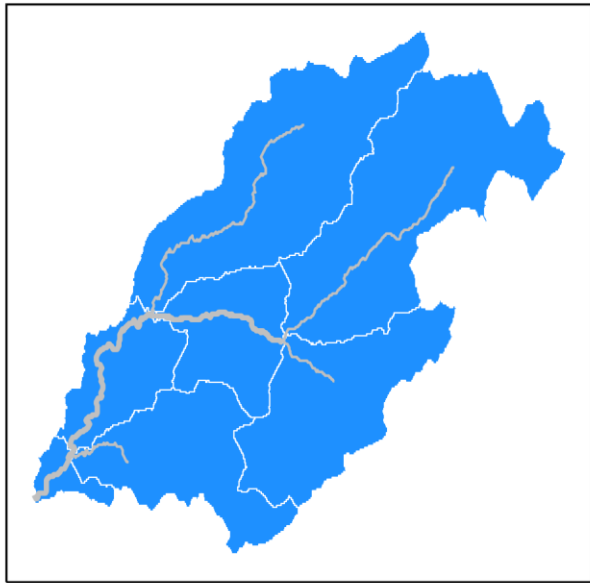
Hydrologic models output runoff, not streamflow

We need **route** the runoff and **generate** streamflow.

Unit hydrograph theory converts runoff values for a basin into flow or discharge at the basin outlet.

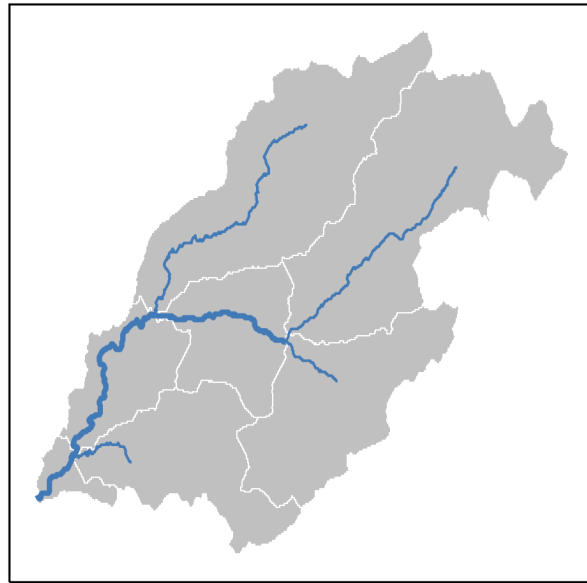


Hydrologic models output runoff...

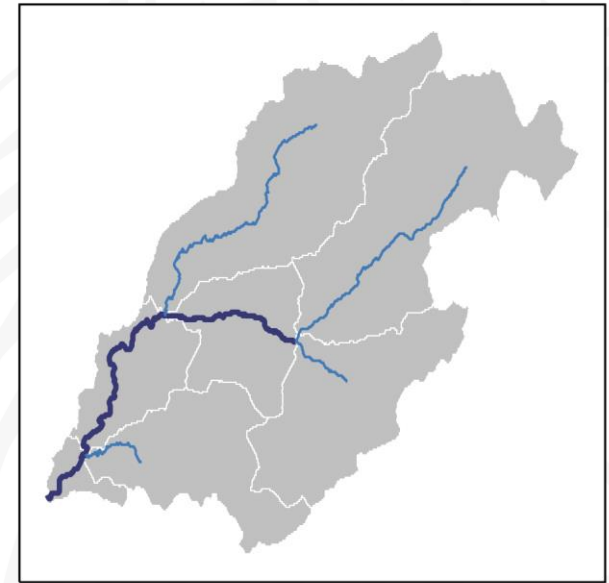


Hillslope routing

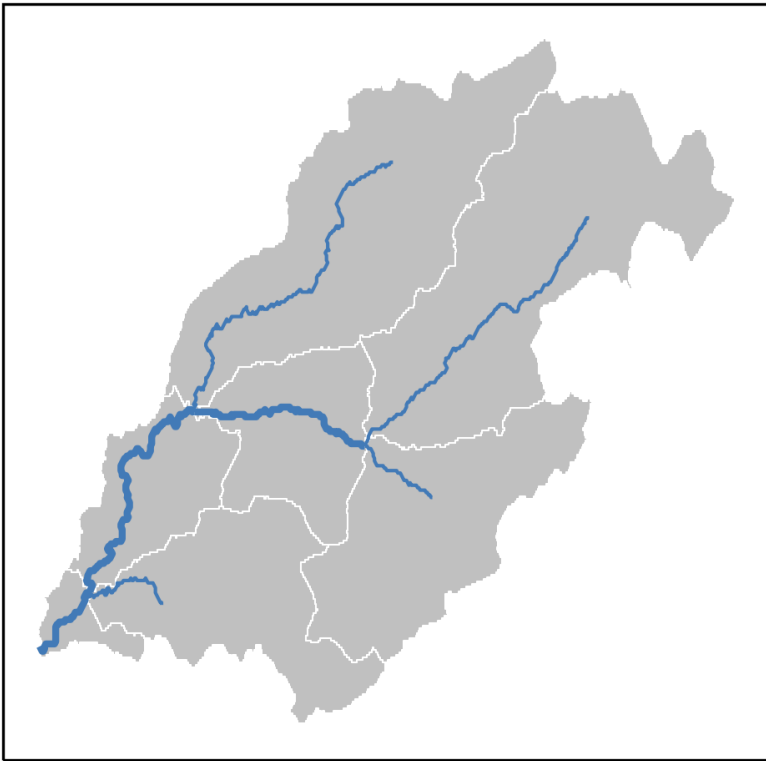
Hillslope routing is performed with a gamma-distribution-based unit-hydrograph to transport runoff from a hillslope to a catchment outlet



River channel routing



Impulse response function – unit hydrograph



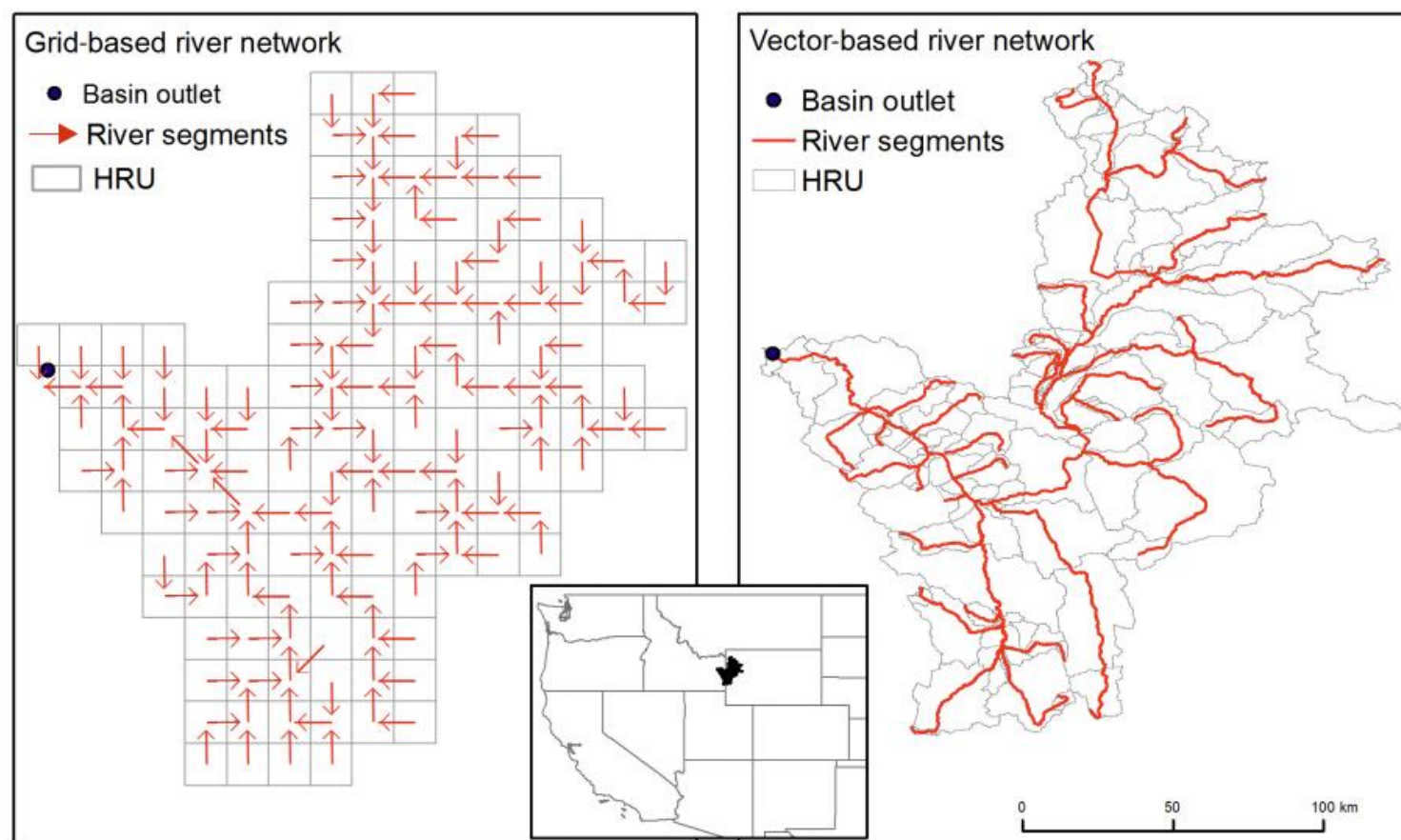
Each pair of the upstream and downstream HRUs has one unit hydrograph

Kinematic wave tracking (KWT)

KWT method computes a wave speed or a celerity for the runoff (or discharge) that enters an individual stream segment from the corresponding HRU at each time step using kinematic approximation

- The runoff, represented as a particle, is propagated through the river network based on a travel time (the celerity divided by the segment length). Note that the wave celerity differs from the flow velocity, as the wave typically moves faster than water mass
- The wave celerity [LT^{-1}] is a function of channel width [L], Manning's coefficient $[-]$, channel slope [unitless], and discharge [L^3T^{-1}].

Grid-based v.s. vector-based river networks



How do we run mizuRoute?

1

Download source code

- GitHub

2

Compile source code

- Slightly more work

3

Download example dataset

- Here we download the data from google drive

4

Prepare configuration file

- Relative path or absolute path?

5

Run the model!

Step 1: Download the source code

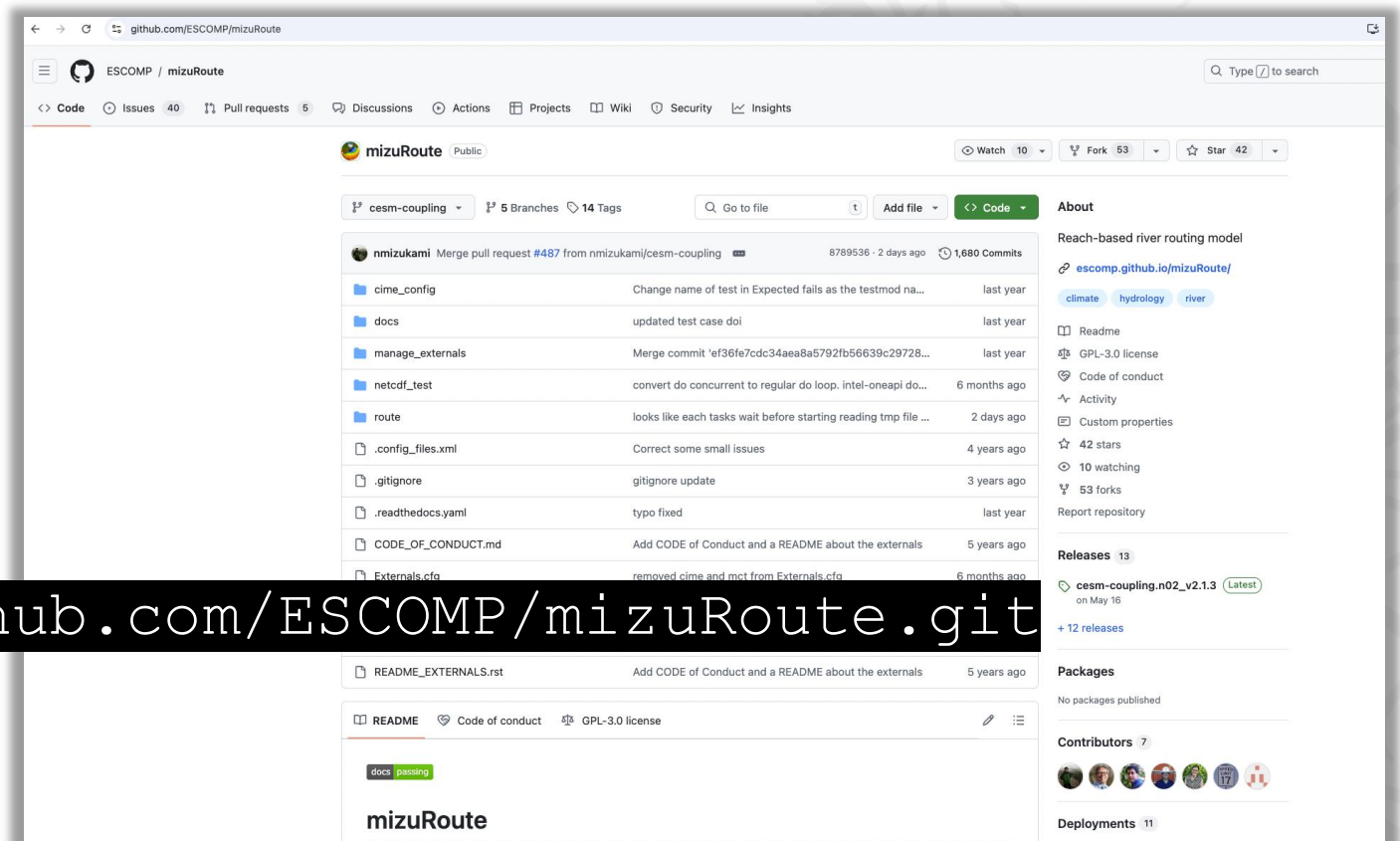
1.1. Go to model directory, type following command to download the mizuRoute model source code in “model” folder, and checkout the “main” branch

```
$ cd ~/model
```

```
$ git clone https://github.com/ESCOMP/mizuRoute.git
```

```
$ cd mizuRoute
```

```
$ git checkout main
```



Step 2: Compile source code

2

Compile source code

- Correct path to the NetCDF library

Step 2: Compile source code

- 2.1. Edit the **Makefile**

```
cd ~/model/mizuRoute/route/build
vim Makefile
```

```
#=====
# User configure part
#=====
# Define fortran compiler - gnu, intel or pgi
FC = gnu

# Define the compiler exe, e.g., gnu=>gfortran, intel=>ifort, pgi=>pgf90
FC_EXE = gfortran

# Define the executable
EXE = route.exe

# Define optional setting
# fast:      Enables optimizations
# debug:     Minimum debug options, still
# profile:   Enables profiling
MODE = debug

# define open MP option (put yes to activate OMP)
isOpenMP =

# Define core directory below which everything resides
# parent directory of the 'build' directory
# do not put space at the end of path
F_MASTER = /home/jovyan/data/model/mizuRoute/route/

# Define the NetCDF libraries and path to include files
ifeq "$(FC)" "gnu"
    NCDF_PATH = /usr/local
else ifeq "$(FC)" "intel"
    NCDF_PATH =
else ifeq "$(FC)" "pgi"
    NCDF_PATH =
```

Step 2: Compile source code

- 2.2. Compile the code, and check whether the executable file is successfully generated!

```
cd ~/model/mizuRoute/route/bin
```

```
make > compile.log
```

```
ls
```

\$ _ jovyan@jupyter-yifancheng X

\$ _ jovyan@jupyter-yifancheng X

(base) jovyan@jupyter-yifancheng-5fub:~\$ cd ~/model/mizuRoute/route/bin

(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route/bin\$ ls

README.md route.exe

Download
example
dataset

3

Download
example dataset

- We can use GitHub to download it as well

Step 3: Download example data

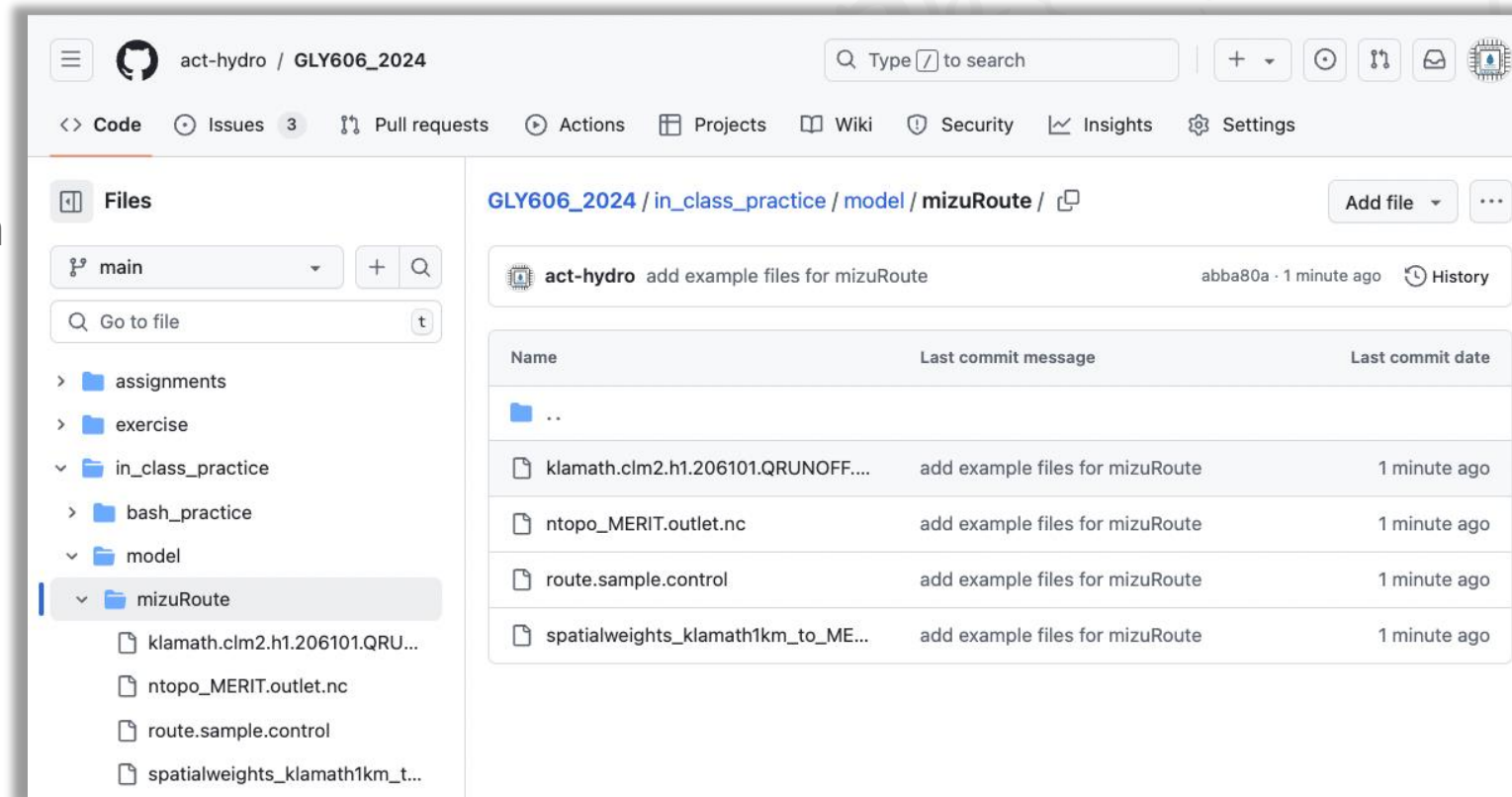
- 3.1. Prepare the folder for the data

```
cd ~/model/mizuRoute/route  
mkdir output
```

```
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route$ mkdir output  
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route$ ls  
ancillary_data  bin  build  input  output  settings
```

Step 3: Download example data

- 3.2. Download the example dataset from
“GLY606_2024/in_class_practice/model/mizuRoute”



The screenshot shows the GitHub interface for the repository 'act-hydro / GLY606_2024'. The left sidebar displays the file tree with the path 'in_class_practice / model / mizuRoute' selected. The main content area shows the commit history for this path, with a table listing the files added in the most recent commit.

Name	Last commit message	Last commit date
..		
klamath.clm2.h1.206101.QRUNOFF....	add example files for mizuRoute	1 minute ago
ntopo_MERIT.outlet.nc	add example files for mizuRoute	1 minute ago
route.sample.control	add example files for mizuRoute	1 minute ago
spatialweights_klamath1km_to_ME...	add example files for mizuRoute	1 minute ago

Step 3: Download example data

- 3.3. Drag files to corresponding folders!

Folder path: / ... / route / input /

Name	Last Modified
klamath.clm2.h1.206101.QRUNOFF.nc	11 minutes ago
README.md	4 hours ago

Folder path: / ... / route / ancillary_data /

Name	Last Modified
ntopo_MERIT.outlet.nc	3 hours ago
param.nml.default	4 hours ago
README.md	4 hours ago
spatialweights_klamath1km_to_MERIT-Hydro.nc	3 hours ago

Folder path: / ... / route / settings /

Name	Last Modified
README.md	4 hours ago
route.sample.control	13 minutes ago
SAMPLE.control	3 hours ago

Step 4: Prepare configuration file



Prepare configuration file

- Relative path or absolute path?

Step 4: Prepare configuration file

4.1. Let's first take a look at the configuration file!

```

$ cd ~/model/mizuRoute/route/settings/
$ cp route.sample.control route.control.klamath
$ vim route.control.klamath
    
```

```

! *****
! *****
! DEFINE DIRECTORIES
! -----
<ancil_dir>          PLACEHOLDER_ANCILLARY_DIR
<input_dir>         PLACEHOLDER_RUNOFF_DATA_DIR
<output_dir>        PLACEHOLDER_OUTPUT_DIR
    
```

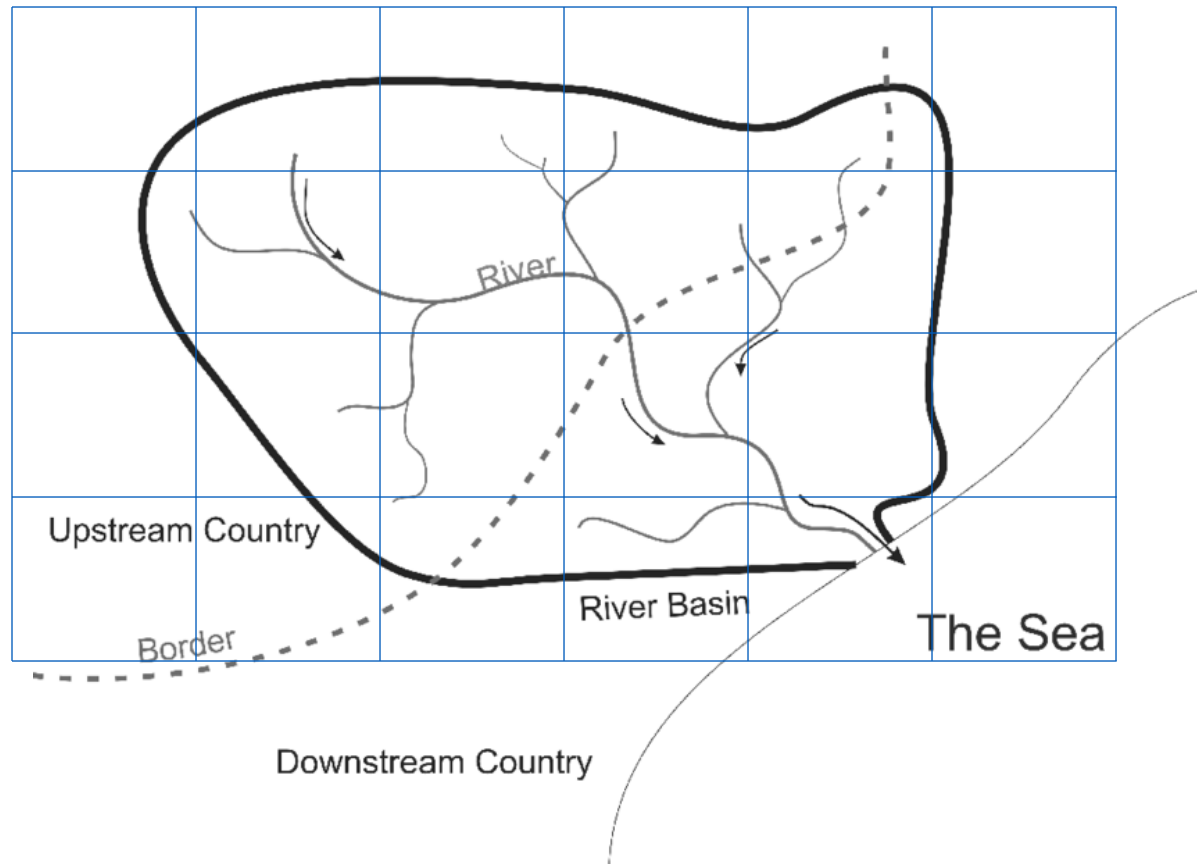
River direction, river length, river properties...

```

! *****
! *****
! DEFINE FINE NAME AND DIM
! -----
<fname_ntop0ld>      PLACEHOLDER_RIVER_TOPO_FILE
<lname_sseg>         seg
<lname_nhru>         hru

! *****
! *****
! DEFINE RUNOFF MAPPING FILE
! -----
<is_remap>           T
to river network HRU
<fname_remap>        PLACEHOLDER_SPATIAL_WEIGHT_FILE
<vname_hruid_in_remap> polyid
mapping file
<vname_weight>       weight
river network HRU
<vname_num_qhru>     overlaps
thin each river network HRU
<vname_i_index>      i_index
<vname_j_index>      j_index
<lname_hru_remap>    polyid
<lname_data_remap>   data
    
```

Step 4: Prepare configuration file



$$Q_{hru} = \sum_{i=1}^n (\%A)_i \times Q_i$$

Step 4: Prepare configuration file

4.1. Let's first take a look at the configuration file!

```

$ cd ~/model/mizuRoute/route/settings/
$ cp route.sample.control route.control.klamath
$ vim route.control.klamath
    
```

```

! *****
*****
! DEFINE DIRECTORIES
! -----
<ancil_dir>          PLACEHOLDER ANCTILIARY DTR
<input_dir>         PLACEHOLDER RUNOFF DATA DIR
<output_dir>        PLACEHOLDER_OUTPUT_DIR
    
```

```

*****
*****
! DEFINE RUNOFF FILE
! -----
<fname_qsim>        PLACEHOLDER INPUT NCFILE
<vname_qsim>        QRUNOFF
<lname_xlon>        lon
<lname_ylat>        lat
<vname_time>        time
<lname_time>        time
<units_qsim>        mm/s
!<calendar>         noleap
<dt_ro>             86400
    
```

Step 4: Prepare configuration file

4.1. Let's first take a look at the configuration file!

```

$ cd ~/model/mizuRoute/route/settings/
$ cp route.sample.control route.control.klamath
$ vim route.control.klamath
    
```

```

! *****
*****
! RUN CONTROL
! -----
<case_name>          klamath
<sim_start>          2061-01-01 00:00:00
<sim_end>            2061-01-31 00:00:00
<restart_write>       never
ed (need to specify date with <restart_date>
!<restart_date>      2061-01-30 00:00:00      ! restart date.
!<fname_state_in>    clm50.mizuRoute.r.2007-12-31-00000.nc
<newFileFrequency>   yearly
<dt_qsim>            86400
    
```

```

! *****
*****
! ROUTE options
! -----
<route_opt>          3
WT, 3->KWE
<doesBasinRoute>     1
! *****
*****
! output options
! -----
<basRunoff>          T
<instRunoff>         F
<dlayRunoff>         T
<sumUpstreamRunoff>  T
<KWTroudedRunoff>    T
<IRFroudedRunoff>    T
<MCroudedRunoff>     T
<KWroudedRunoff>     T
<DWroudedRunoff>     T
    
```

Practice: Complete the configuration file

- Fill in the directories (PLACEHOLDER_XXX_DIR) and filenames (PLACEHOLDER_XXX_FILE)

```
$ vim route.control.klamath
```

```
! ****
*****
! DEFINE DIRECTORIES
! -----
<ancil_dir>          PLACEHOLDER_ANCILLARY_DIR
<input_dir>         PLACEHOLDER_RUNOFF_DATA_DIR
<output_dir>        PLACEHOLDER_OUTPUT_DIR
```

```
! ****
*****
! DEFINE RUNOFF FILE
! -----
<fname_qsim>         PLACEHOLDER_INPUT_NCFILE
<vname_qsim>        QRUNOFF
```

```
! ****
*****
! DEFINE FINE NAME AND DIMENSIONS
! -----
<fname_ntop0ld>      PLACEHOLDER_RIVER_TOPO_FILE
<lname_sseg>         seg
<lname_nhru>         hru
```

```
! ****
*****
! DEFINE RUNOFF MAPPING FILE
! -----
<is_remap>           T
  to river network HRU
<fname_remap>        PLACEHOLDER_SPATIAL_WEIGHT_FILE
```

Two terminal methods

```

! *****
*****
! ***** DEFINITION OF MODEL CONTROL INFORMATION *****
*****
! *****
*****
! *****
*****
! Note: lines starting with "!" are treated as comment lines — there is no limit on the number of comment lines.
!   lines starting with <xxx> are read till "!"
!
! *****
*****
! DEFINE DIRECTORIES
! _____
<ancil_dir>                /home/jovyan/data/model/mizuRoute/route/ancillary_data/
! directory containing ancillary data (river segment data,
<input_dir>                /home/jovyan/data/model/mizuRoute/route/input/
! directory containing input data
<output_dir>              /home/jovyan/data/model/mizuRoute/route/output/
! directory containing output data
! *****
*****
! RUN CONTROL
! _____
<case_name>                klamath                                ! simulat
ion name - used for output netcdf name
<sim_start>                2061-01-01 00:00:00                  ! time of
simulation start (year-month-day)
<sim_end>                  2061-01-31 00:00:00                  ! time of
simulation end (year-month-day)
<restart_write>            never                                ! restart w
rite option. never, last, Annual, specified (need to specify date with <restart
_date>
!<restart_date>            2061-01-30 00:00:00                  ! restart date. activated
only if <restart_write> is "specified"

1,1
Top

```

```

jovyan@jupyter-yifancheng X  jovyan@jupyter-yifancheng X
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route/ancillary_data$ l
s
ntopo_MERIT.outlet.nc  param.nml.default  README.md  spatialweights_klamath1km_
to_MERIT-Hydro.nc
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route/ancillary_data$ p
wd
/home/jovyan/data/model/mizuRoute/route/ancillary_data
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route/ancillary_data$ l
s
ntopo_MERIT.outlet.nc  README.md
param.nml.default      spatialweights_klamath1km_to_MERIT-Hydro.nc
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route/ancillary_data$

```

Practice: Complete the configuration file

- Fill in the directories (PLACEHOLDER_XXX_DIR) and filenames (PLACEHOLDER_XXX_FILE)

```
$ vim route.control.klamath
```

```
! ****
*****
! DEFINE DIRECTORIES
! -----
<ancil_dir>          PLACEHOLDER_ANCILLARY_DIR
<input_dir>         PLACEHOLDER_RUNOFF_DATA_DIR
<output_dir>        PLACEHOLDER_OUTPUT_DIR
```

```
! ****
*****
! DEFINE RUNOFF FILE
! -----
<fname_qsim>         PLACEHOLDER_INPUT_NCFILE
<vname_qsim>        QRUNOFF
```

```
! ****
*****
! DEFINE FINE NAME AND DIMENSIONS
! -----
<fname_ntop0ld>      PLACEHOLDER_RIVER_TOPO_FILE
<lname_sseg>         seg
<lname_nhru>         hru
```

```
! ****
*****
! DEFINE RUNOFF MAPPING FILE
! -----
<is_remap>           T
  to river network HRU
<fname_remap>        PLACEHOLDER_SPATIAL_WEIGHT_FILE
```

```
<ancil_dir>          /home/jovyan/data/model/mizuRoute/route/ancillary_data/
! directory containing ancillary data (river segment data,
<input_dir>          /home/jovyan/data/model/mizuRoute/route/input/
! directory containing input data
<output_dir>         /home/jovyan/data/model/mizuRoute/route/output/
! directory containing output data
```

```
<fname_ntop0ld>      ntopo_MERIT.outlet.nc          ! name of
netCDF containing river segment data
```

```
<fname_qsim>         klamath.clm2.h1.206101.QRUNOFF.nc      ! list of
input netCDF names
```

```
<fname_remap>        spatialweights_klamath1km_to_MERIT-Hydro.nc  ! name of
netCDF containing poly-poly mapping data
```

IT'S
TIME TO
RUN!!!



Step 5: Run the model

5.1. How to run the model!

```
$ cd ~/model/mizuRoute/route  
$ ./bin/route.exe settings/route.control.klamath
```

 jovyan@jupyter-yifancheng X

 jovyan@jupyter-yifancheng X

```
(base) jovyan@jupyter-yifancheng-5fub:~$ cd ~/model/mizuRoute/route  
(base) jovyan@jupyter-yifancheng-5fub:~/model/mizuRoute/route$ ./bin/route.exe  
settings/route.control.klamath
```